

Practical 12 - Development and evaluation of risk prediction models

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

Datasets required: rotterdam_p10.dta

Introduction

In this practical we will perform analyses similar to those shown in the lecture, using the Rotterdam breast cancer data. The aim is to develop a prediction model for 5-year risk of the composite outcome of disease recurrence or death from any cause, and to evaluate the predictive performance of the model.

The data used is the same as in Practicals 10 and 11. The variables we will use are summarised in the table below.

Variable name	Description
pid	Patient identifier
year_cat	Year of surgery. 0: before 1990, 1: 1990 or later
age	Age at surgery
meno	Menopausal status. 0=premenopausal, 1=postmenopausal
size	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50mm
grade	Differentiation grade: categorical with levels 2 and 3
enodes	Number of positive lymph nodes, transformed as $\text{enodes} = \exp(-0.12 * \text{nodes})$
pgr	Progesterone receptors (fmol/l)
er	Estrogen receptors (fmol/l)
chemo	Chemotherapy: 0=no, 1=yes

Variable name	Description
<code>hormon</code>	Hormonal treatment: 0=no, 1=yes
<code>time</code>	Years to the first of recurrence or death without recurrence
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>outcome_cr</code>	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Fit prediction models and evaluate their performance.
- Evaluate prediction model performance using measures of calibration, discrimination, and overall performance.
- Use random split-sampling and k-fold cross-validation for evaluation of prediction model performance.

Question 1

- Load the data.
- It turns out to be sensible to change the scale of the variables `pgr` and `er`, so we will divide each of these by 100 for this analysis.
- Create a training set on which the model will be fitted, and a test set on which the predictive performance of the model will be evaluated. The training set is a random sample of 70% of the cohort, and the remaining 30% for the test set.

Question 2

- Use `stset`.
- Using the training set, fit a Cox regression for the composite outcome using the following predictors: `hormon`, `chemo`, `age`, `meno`, `size`, `grade`, `enodes`, `pgr100`, `er100`, `year_cat`.
- Use the model to obtain estimates of 5-year risk for individuals in the test set.
- What is the range of the predicted risks in the test set?

Question 3

In this question we will look at two measures of calibration: mean calibration and the calibration slope. These are obtained for the test data. The code is given, but you can probably do all of this without looking at it.

Mean calibration

- Calculate the mean estimated 5-year risk in the test data, and compare this with the Kaplan-Meier estimate of risk up to 5 years.
- What is the ratio of the observed to the expected risk at 5 years?
- The solution shows how to obtain a 95% confidence interval for this ratio.

Calibration slope

- Calculate the linear predictor from the risk model for each individual.
- Perform a Cox regression of the outcome on this linear predictor.
- What is the resulting coefficient? This is the calibration slope.

Question 4

Follow the steps below to create a calibration plot:

- Divide the predicted risks in the test data into 10 equal-sized groups (by placing cut-points at the deciles).
- Calculate the mean of the predicted risks for the people in each of the 10 groups.
- Obtain the Kaplan-Meier estimate of 5-year risk in each group.
- Plot the Kaplan-Meier estimates against the mean predicted risks.

Discuss what the plot shows.

Question 5

In this question we create a smooth version of the calibration plot in Question 4. Use the code provided to implement the following steps (all in the test data):

- Calculate $Z_i = \log\{-\log(1 - \widehat{R}_i(5))\}$ for each person in the test data, where $\widehat{R}_i(5)$ is the estimated 5-year risk.
- Perform a Cox regression including Z_i as the only covariate, using a spline (this is done using a restricted cubic spline in the code). This is called the ‘refitted model’.
- Obtain estimates of risk up to 5 years for each person based on the refitted model including Z_i .
- Plot the estimated risks from the refitted model against the estimated risks from the main prediction model.

Question 6

In this question we look at measures of discrimination.

- Use the code provided to obtain estimates of Harrell's c-index.
- Interpret the estimate.

There is no clear way of obtaining Uno's c-index from Stata

Question 7

Lastly we look at the Brier score, which is a measure of overall performance.

- Use the code provided to obtain an estimate of the Brier score and the scaled Brier score.
- Interpret the estimates. What does the scaled Brier score compare?

Question 8

So far we fitted the prediction model in the training data and evaluated the model performance in the test data. A better approach is to use k-fold cross validation. As a reminder, this involves the following steps, for 10-fold cross-validation:

- Divide the full data (i.e. before it was split in to training and test sets) randomly into approximately 10 equal-sized groups ('folds').
- Exclude fold 1 and fit the model in folds 2-10. Then obtain risk predictions in fold 1.
- Exclude fold 2 and fit the model in the remaining 9 folds. Then obtain predictions in fold 2.
- ...and so on, excluding each fold one by one. This results in a cross-validated risk prediction for each individual.
- Measures of predictive performance are then obtained using the cross-validated risk predictions on the full data.

See the code for an example of using 10-fold cross-validation.

Question 9

- Suppose you were happy to proceed with your model on the basis of the above evaluations of predictive performance. Obtain and write down a formula that could be used to obtain a prediction for a new individual.